

RESEARCH ARTICLE

Power to the People! Assigning Popular Music Genres Based on Listeners' Perspective Using a Genre Tree Approach

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Abstract

Despite popularity of playlists on streaming platforms, music genres still remain a primary organizational tool among listeners, stakeholders, and researchers. Yet both 'correct' genre labels for popular music exemplars and structure of genre taxonomies are often contested. Nevertheless, questionable categorical genre assignments form the basis of many music genre recognition models in MIR and recommender systems research as well as in musicological basic research on music preferences and functions, raising concerns about reliability, validity, and comparability of results. In contrast, popular music studies have long emphasized the listener-dependent nature of genre assignments. In other words: Genre assignment may be considered a matter of majority voting within a listener community. So why not make a virtue of necessity? We present a new methodology for constructing hierarchical genre trees from *MusicBrainz* folksonomy genre votes and demonstrate feasibility of this approach for 167,762 tracks listened to in Germany across the past 50 years. The resulting Community Music Genre Tree for popular music in Germany (COMGET-G) comprises 234 genres—interesting for cultural analyses, but probably too detailed for many applied research tasks. We address this by presenting an additional algorithm for folding such trees to a desired number of genre categories for any research problem (FOLDCAT). For automatic category assignment of music tracks still lacking community votes, we also develop ML classifiers at multiple granularity levels using openly-available music features and metadata from different online databases. Exemplary results indicate that our approach is able to assign a proper genre category representing a multiple listeners' perspective to any popular music track in a transparent, reproducible, and intersubjectively comprehensible way.

Keywords: music genre recognition, hierarchical taxonomy, folksonomy, network analysis, machine learning classification, German popular music

1. Introduction

Although playlist-driven listening has grown alongside the global spread of streaming platforms, genre labels still remain the primary way people and institutions organize popular music: Listeners search and browse by genre; playlist editors and DJs assemble genre playlists; A&R, marketing and label teams position and promote artists using genre tags; and music journalists and researchers rely on genre labels to describe and compare popular music (Bonini and Magadda, 2024). Some scholars argue that classificatory practices have weakened since the late twentieth century (DiMaggio, 1987) and that certain genre boundaries are unstable (Silver et al., 2016), but empirical evidence shows that listening and social signaling re-

main organized around genres (Airolti et al., 2016; Lonsdale, 2021). On most streaming platforms, genre assignments are embedded in metadata and pipeline logics—and hence used as search filters, playlist seeds, and features in recommender systems. Thus, platform algorithms amplify genres' practical significance for discovery and market positioning (Valverde, 2025). For researchers, conversely, genre labels serve both theoretical and methodological roles: they form variables in preference and reception studies (anonymized authors, XXXX), inputs to recommender-system research (Porcaro et al., 2024), and useful aggregator variables for measuring perceived diversity and mitigating hubness in high-dimensional audio and metadata feature spaces (Flexer et al., 2018; Flexer and

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Grill, 2016).

However, ‘correct’ genre labels for individual popular music phenomena (i.e., tracks, artists, releases, scenes, market, playlists, labels, performances etc.) are frequently contested (van Venrooij and Schmutz, 2018). Different actors may attach different genre labels to music releases for different reasons: artists employ them to signify their musical identity; publishers and distributors tag releases to support market placement; critics and musicologists apply normative or historical taxonomies; and listeners use genre terms to organize their listening collections and to negotiate their distinctive musical identity in everyday communicative interactions (Frith, 1996; Negus, 1999). Not only do chosen genre labels diverge, but the taxonomical systems employed also differ strongly in degree of differentiation. Commercial platform taxonomies illustrate this: For example, the commercial music metadata platform *Gracenote* uses a compact set of 25 genre tags (Jiang et al., 2024), whereas *Spotify* exposes many thousands of genre tags (McDonald, nd). Music databases and platforms also differ in terms of the objects to which they assign genres (track, release, or artist) and in assignment strategies (single primary genre versus multi-label tag sets).

These differences pose methodological challenges for empirical researchers across MIR and computational musicology. Problems include inconsistent genre labels that may degrade generalizability of supervised machine-learning models (Bogdanov et al., 2019), survey items on musical preferences that might be interpreted very heterogeneously by different listeners (Lange et al., 2025), and, on a larger scale, the issue of cross-study noncomparability that undermines replication and meta-analysis (Sturm and Flexer, 2023). Faced with these constraints, researchers often pragmatically adopt available commercial or open taxonomies, without systematically engaging with the actual source of labels or underlying theoretical assumptions (Green et al., 2024). To address these challenges, the present paper presents a reproducible procedure to derive a scalable and transparently constructed genre taxonomy for any given corpus of popular music based on the community votes of any genre folksonomy. Paired with algorithmic music-genre recognition, this approach can (1) surface empirical regularities and disagreements in genre label use that may inform popular music studies, and (2) provide a shared, probabilistic ‘currency’ for cross-disciplinary work in computational musicology and MIR—so that classifiers, preference studies, and sociological analyses may operate from a common, well-documented mapping between recordings and meta-genres.

1.1 Musical Genres as hierarchical discourse formation

In popular music, genres are organized hierarchically: Pachet and Cazaly (2000) identified genealogical links

(e.g., *rock* as parent of *alternative rock*), geographical links (*hip hop* as supergenre of *German hip hop*), and historical-period links (*pop* as supergenre of *80s pop*) in existing taxonomies. As Brackett (2016, p. 8) argues, hierarchy depth is not limited to two levels but can extend virtually indefinitely—yet consensus on which labels to select and how to relate them remains contested.

Expert-constructed taxonomies show limited agreement (Pachet and Cazaly, 2000), and expert classifications often diverge from listener community norms (Sordo, 2008). Adopting the power-critical perspective of cultural studies (du Gay et al., 2013), we interpret the discourse on genres in the popular music field through Foucault’s (2013) concept of *discourse formation*. From this viewpoint, artists, industry, and listeners exert unequal influence over which genre labels circulate and stabilize over the long run (Lena and Peterson, 2008). Industry actors’ genre assignments capture how the music industry positions and markets artists within commercial taxonomies, which may diverge from listener perception due to strategic positioning, legacy labelings, or cross-genre appeal (Valverde, 2025). We term these industry-driven assignments *distribution genres*. By contrast, aggregating listener votes treats genre assignment as implicit majority voting within a community, privileging the perspective of those who engage with the music rather than those who distribute it. This democratic aggregation replaces single-source genre assignments from label A&R departments, artists, platform editors, or individual experts with collective consensus, capturing how listeners actually categorize music in everyday discourse. We term these listener-derived assignments *community music genres*, reflecting the emergent consensus of listening communities rather than the view of economically powerful institutional gatekeepers.

To empirically ground musical genres for research in listener discourse rather than institutional authority, we propose a new methodological approach that derives a *community music genre* hierarchy and *community genre* assignments for music tracks from the votings of global music listener communities, also termed *folksonomies*. Platforms such as *RateYourMusic* (nd), *Discogs* (2024), or *MusicBrainz* (nd) provide crowd-sourced ‘ground truth’ on the genre of musical objects reflected in aggregated tagging behavior. *MusicBrainz* seems particularly suitable for our endeavor: it maintains a detailed whitelist of over 2,000 genre tags without imposing an a-priori hierarchy, and its democratic voting system allows users to assign and vote on tags at multiple levels (track, release, artist), enabling nuanced analysis of consensus and disagreement.

1.2 Application of Genres in Empirical Music Research

For the application of hierarchical genre taxonomies in MIR research and computational musicology, several

pragmatic issues arise. First, balanced genre distributions improve statistical power and machine learning performance (Japkowicz and Stephen, 2002). However, subgenres typically receive fewer track assignments than their super categories (Bogdanov et al., 2019), creating inherent imbalance that propagates through hierarchical levels. A method to empirically derive hierarchical genre taxonomies should therefore optimize balance within each hierarchy level separately, ensuring well-structured trees where sibling genres exhibit comparable sizes regardless of their position in the hierarchy.

Second, different research questions demand different numbers of *genre categories* that combine genres within the hierarchy into broader groupings, allowing for different *granularities*. Granularity refers to the level of detail at which genres are differentiated: low granularity collapses many subgenres into few broad categories (e.g., *rock*, *pop*, *electronic*), whereas high granularity preserves fine-grained distinctions (e.g., *death metal*, *synthwave*, *UK garage*). Surveys of musical taste may require moderate to high granularity (Lange et al., 2025, e.g., 24 genres), machine learning models often perform better with fewer, broader categories (Genussov and Cohen, 2010), and subcultural analyses may demand fine-grained subgenre differentiation (Caparrini et al., 2020). Preserving consistency with the overall tree structure remains desirable across granularities. This requires a method to derive genre categories at multiple scales from a single underlying hierarchy.

Third, derived genre categories should demonstrate validity and generalizability to unseen popular music items. External validation against other genre assignments commonly used in MIR and computational musicology research tests whether categories reflect broader consensus beyond a single platform (Sturm and Flexer, 2023). Additionally, automated classifiers trained on these categories serve two purposes: they assigns genres to popular music objects lacking community labels, and evaluate whether the categories capture systematic patterns learnable from musical and contextual features.

1.3 Feature dimensions for music genre recognition

Automated classification of musical genres, or music genre recognition, is a core task in Music Information Retrieval. Green et al. (2024) provide a systematic state-of-the-art review and critique how genre theory from the social sciences and humanities is often overlooked when selecting features (for a systematic review on genre theories see Merlini, 2020). Researchers in MIR and music psychology commonly assume that genre can be inferred from audio features alone (Tzanetakis and Cook, 2002; Rentfrow et al., 2011). This conflicts with a sociological approach to genres, which we adopt here, describing

them as “systems of orientations, expectations and conventions that circulate between industry, text and subject” (Neale, 1980, p. 19), emphasizing extra-musical context alongside sound. Franco Fabbri’s (1982) genre theory offers such a synthesis: musical style (instrumentation, rhythm, harmony, timbre) and release milieu (subcultural context, artist positioning, market placement) together inform how listeners assign genres. Modern MIR tools extract style features from audio signals (Bogdanov et al., 2013), and open music databases provide distributional metadata (label, chart performance, platform placement).

Lyrics occupy an intermediate position, conveying emotional expression and stylistic content (Brand et al., 2019) as well as contextual information through language codes and topics that signal subcultural affiliation (Carbone et al., 2024). A classifier might need to pay special attention to the *mainstream pop* or *pop* metagenre that Fabbri (1982) describes as encompassing eclectic, commercially optimized works that resist association with a specific milieu. Artists from former niche scenes are often retrospectively relabeled as *pop* or *mainstream* after achieving commercial success (Holt, 2007, p. 20). Therefore, popularity itself appears to be a relevant feature dimension for classification. The anonymized authors (XXXX) recently proposed a feature space spanning four dimensions—musical style, release context, lyrical content, and popularity to operationalize the multi-dimensional constitution of genre following Fabbri’s (1982) approach.

Drawing on the theoretical and methodological foundations outlined above, the present work pursued three research questions:

RQ1 How can we construct an empirically grounded and sufficiently balanced hierarchical genre tree from multi-tag folksonomy votes for a given popular music corpus?

RQ2 How can we aggregate the inferred tree structure to a scalable number of distinct genre categories suitable for machine learning and statistical applications?

RQ3 How valid and generalizable are community-derived genre category assignments for empirical music research?

a) To what extent do the community-derived categories agree with alternative folksonomy taxonomies (e.g., Discogs)?

b) Can supervised classifiers reliably predict community-derived genre categories at multiple granularities using stylistic, release-milieu, lyrical, and popularity features?

In the following sections, we will hence introduce a method to create a balanced hierarchical genre taxonomy based on genre co-occurrences in folksonomy tags, describe a tree folding algorithm to flexibly identify suitable genre categories at different levels of gran-

ularity for various research tasks, present automatic classifiers for different numbers of genre categories, and finally discuss our findings and summarize the results.

2. RQ1: Balanced Hierarchical Genre Tree

2.1 Data – Popular Tracks in Germany (POPTRAG)

As initial demonstration of our new methodological approach, we perform the whole analysis depicted in this paper with a corpus of 254,334 unique popular tracks released and frequently listened to in Germany over the past 50 years (POPTRAG). Four sources comprise this corpus: 215,702 tracks from weekly *MediaControl* German single and album charts (18 Nov 1977–21 Nov 2024; reduced to 75% where linkage success to *Spotify* identifiers was possible); 26,949 *Spotify* recommendations from 126 German-market genre seeds (retrieved 29 Nov 2024; median 325 per seed, IQR 25, range 49–343); 8,801 tracks from 100 top-featured *Spotify* playlists for the German market (retrieved on 29 Nov 2024, median 90 tracks per playlist, IQR 42, range 28–750); and, finally, 11,994 entries from *Germany’s Daily Top 200 Spotify* charts (01 Jan 2017–30 Nov 2024). From all sources, we excluded tracks shorter than 30 seconds (pertains to 853 tracks, likely non-musical snippets) and retained only unique title-artist combinations, preferring earlier release dates for duplicates.

In a second step, we then tried to acquire *MusicBrainz* track genre tags for all music tracks in the POPTRAG corpus. Hereto, only genre labels contained the *MusicBrainz* whitelist (snapshot 16 Jun 2025) were taken into account, reduced by tags referring to genres beyond popular music (*non-music*, *interview*, *comedy*, *nature sounds*, *children’s music*, *spoken word*, *standup comedy*). In case track-level tags were missing, we employed release tags instead. If these were missing, too, we used the artist tags. In addition, we required any genre tag to appear with at least 50 distinct artists in the overall corpus to ensure sufficient discursive support for any genre, if not, they were discarded. Finally, we retained only tracks in the corpus that were exhibiting at least two community genre assignments fulfilling all of these criteria, leading to a removal of 86,613 tracks (34%) and yielding a *MusicBrainz*-tagged subset of $N = 167,721$ tracks annotated with 234 distinct music genre tags spanning release years 1924–2024 (median 2012, IQR 16 years). Community genre assignments comprised between 1 and 22 different tags per track (median 4, IQR 3) and cast between 2 and 135 votes per track (median 6, IQR 7).

2.2 Procedure: From Genre-Tag Co-Occurrences to an Hierarchical Genre Tree

Based on the denoised German POPTRAG corpus, we algorithmically built a hierarchical genre tree from *MusicBrainz* tag co-occurrences, adapting the method pre-

sented by Schreiber (2015; 2016). Thereto, we first constructed a weighted co-occurrence network where nodes represented *genre* labels and directed edges $e_{A \rightarrow B}$ quantified how strongly *genre A* predicted *genre B* on the same track. For each ordered pair (A, B) , we computed a vote-weighted conditional frequency:

$$e_{A \rightarrow B} = P(B | A) \times \frac{1}{|T_{A,B}|} \sum_{t \in T_{A,B}} \frac{v_{t,B}}{v_{t,\text{total}}} \quad (1)$$

where $T_{A,B}$ denotes tracks tagged A and B , $v_{t,B}$ the votes for B on track t , and $v_{t,\text{total}}$ the total votes on t . This formula produced asymmetric edges ($e_{A \rightarrow B} \neq e_{B \rightarrow A}$) that combined directional co-occurrence with community vote strength. To illustrate: suppose 1,000 tracks carried *metal* and 600 also carried *rock*, giving $P(\text{rock}|\text{metal}) = 0.6$. If the average vote share for *rock* on co-tagged tracks equalled 0.5, then $e_{\text{metal} \rightarrow \text{rock}} = 0.6 \times 0.5 = 0.3$. Conversely, 6,000 tracks carried *rock*, 600 of those carried *metal*, so $P(\text{metal}|\text{rock}) = 0.1$; with an average *metal* vote share of 0.7 on *rock* tracks, $e_{\text{rock} \rightarrow \text{metal}} = 0.07$. These weights express both prevalence and tag confidence.

In a second step, we resolved reciprocal track connections to obtain a unidirectional scaffold. For each pair, we thereto subtracted the weaker directional probability from the stronger and kept only the positive residual:

$$e'_{A \rightarrow B} = (e_{A \rightarrow B} - e_{B \rightarrow A})_+ \quad (2)$$

$$e'_{B \rightarrow A} = (e_{B \rightarrow A} - e_{A \rightarrow B})_+ \quad (3)$$

In the example this yields $e'_{\text{metal} \rightarrow \text{rock}} = 0.6 - 0.07 = 0.53$ and $e'_{\text{rock} \rightarrow \text{metal}} = 0$, making *metal* $\rightarrow$ *rock* a clear candidate supergenre link.

Third, we enforced a strict tree-like structure by assigning each node its single strongest outgoing edge as its *supergenre*; all other edges were discarded. Components without *supergenre* were attached to an artificial root *POPULAR MUSIC* to guarantee connectivity. In the example, *metal* would attach to *rock* unless a stronger edge existed, and *rock* would attach to the root if it had no *supergenre*.

Finally, we balanced the tree in terms of optimizing leaf-node populations on each tree level. Thereto, we estimated each genre node’s size as the sum of its assignment probabilities across tracks in the corpus, then computed the Gini coefficient within each hierarchical tree-level for all genre nodes contained, and defined overall tree imbalance as the weighted mean of all tree-level Ginis. We then searched for the least-imbalanced tree by iteratively removing the weakest hierarchical link of the whole tree, reattaching the resulting dangling tree to *POPULAR MUSIC*, and recomputing imbalance again; we then finally selected the iteration with minimal overall tree imbalance.

2.3 Results: Community Music Genre Tree for Popular Music in Germany (COMGET-G)

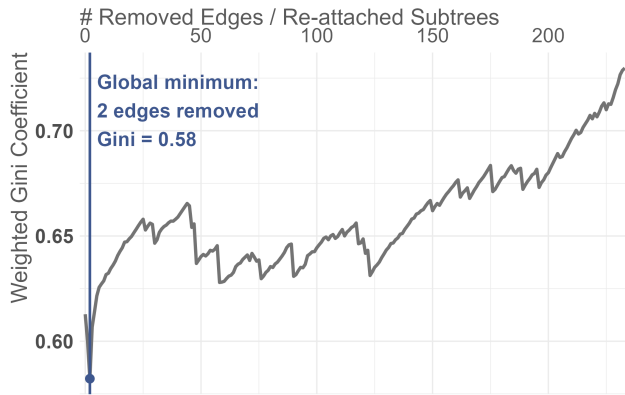


Figure 1: Weighted Gini coefficient (0–1) across iterations of cutting weak genre links and reconnecting dangling subtrees to *POPULAR MUSIC*.

Level	# nodes	# tracks	Gini
1	3	total: 47,846 (29%) 12,472; 16,401; and 18,973	0.136
2	119	total: 90,381 (54%) median: 263 IQR: 789 range: 10.4 to 5832	0.643
3	110	total: 29,343 (17%) median: 156 IQR: 259 range: 14.7 to 1496	0.531
4	2	total: 192 (0.1%) 55.6 and 136	0.420

Table 1: Structure of COMGET-G at different hierarchy levels; track counts are sums of genre tag assignment probabilities in POPTRAG.

The initial POPTRAG *genre* co-occurrence network contained 234 *genre* nodes forming a single connected component with 45.14% density, indicating relatively high inter-tag connectivity. By median, *genres* co-occurred with 97 different others (IQR 63.5; range 3 for *waltz* to 229 for *pop*), revealing large differences between niche and ubiquitous tags. After resolving reciprocal directions, each *genre* received on average 29.5 incoming subgenre links (IQR 32; min 0; max 228 for *pop rock*) and 55 outgoing supergenre links (IQR 32; min 0 for *rock*; max 105 for *dub*).

The balance optimization procedure (Figure 1) reattached *pop* and *hip hop* directly to *POPULAR MUSIC*—*pop* had formerly been subordinate to *rock*, and *hip hop* to *pop*, however these connections were the weakest in the whole tree and their cutting resulted in clear improvment of overall balance. The three weakest remaining links are *electronic* to *pop* (0.22), *classical* to *pop* (0.27), and *jazz* to *pop* (0.29). The final Com-

munity Music Genre Tree for popular music in Germany (COMGET-G) is highly leaf-heavy: 89.36% of genre-nodes lack subgenres. *Rock* emerges as the largest *supergenre* with 63 direct *subgenres*, and only *rock*, *pop*, and *hip hop* remain connected directly to *POPULAR MUSIC*, forming three broad meta-categories. The tree exhibits four hierarchy levels (excluding the artificial root); see Table 1 for structural details. An interactive visualization of COMGET-G is available in the digital appendix.

3. RQ2: Scalable Granularity

3.1 Methods: Folding Algorithm (FOLDCAT)

To derive a scalable number of broader *genre categories* suitable for diverse research questions and statistical applications from COMGET-G, we developed a greedy folding algorithm (FOLDCAT) with three objectives: optimal *genre category* balance, a user-selectable number of *genre categories*, and sufficient track counts per *genre* for robust machine learning and statistical analysis. FOLDCAT iteratively processes leaf *genres* beginning with those deepest in the hierarchy and largest in track count, continuing to the root. For each leaf *genre*, the algorithm checks whether it meets a minimum track count threshold. If it does and its supergenre can sustain removal, the algorithm keeps that leaf *genre* as a *genre category*. Otherwise, the algorithm absorbs the leaf *genre*—along with its siblings and any descendants already processed—into its supergenre by remapping all track assignments upward. This procedure continues until all remaining *genres* meet the minimum threshold.

Here, *genre size* is again computed as the sum of vote-weighted assignment probabilities across all tracks. For a desired number of *genre categories*, users select the folding solution corresponding to a local Gini minimum, drawing on a hierarchy-level-weighted Gini coefficient (see section 2.2), thereby ensuring well-balanced *categories* across all hierarchy levels. A detailed flowchart of this procedure appears in the appendix. For the present exemplary analysis using the POPTRAG corpus, we conducted a grid search across minimum *genre sizes* from 1,000 to 15,000 in increments of 5, then selected four different local Gini minima from the results for comparison.

3.2 Results: Germany's Popular Music Genre Categories

For our POPTRAG corpus, the folding algorithm identified multiple optimal solutions for a COMGET-G (Figure 2): a low-granularity solution with 5 *genre categories* (min. 12,930 tracks, Gini 0.307), medium-granularity solutions with 10 and 12 *genre categories* (min. 4,100 and 3,520 tracks; Gini 0.343 and 0.342), a high-granularity solution with 25 *genre categories* (min. 1,485 tracks, Gini 0.367), and solutions with higher granularity, comprising 32, 34, and 38 *genre categories*

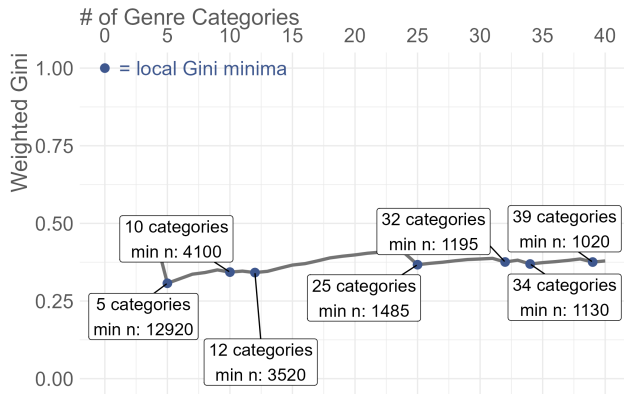


Figure 2: Weighted Gini coefficient (0–1) across COMGET-G folding solutions for minimum track counts from 1,000 to 15,000 in increments of 5, using the POPTRAG corpus.

(min. 1,195, 1,130, 1,020 tracks; Gini 0.376, 0.369, 0.375). We focused subsequent analyses on solutions with 5, 12, 25, and 32 *genre categories*, representing different granularities suitable for various research questions (Figure 3).

COMGET-G5 comprises *rock*, *pop*, *hip hop*, *electronic*, and *metal* (in descending order of track count). In COMGET-G12, *jazz* and *schlager* emerge as further distinct *pop subgenre categories*, while *rock subgenre categories* including *heavy metal*, *alternative rock*, *pop rock*, *hard rock*, and *indie rock* appear at the same level. Notably, listeners do not classify *heavy metal* as a *subgenre category* of *metal* but as a direct *subgenre category* of *rock* on the same hierarchy level as *metal*.

COMGET-G25 introduces further *rock subgenre categories* (*punk*, *progressive rock*, *folk*, *folk rock*, *classic rock*, *blues rock*, *singer-songwriter*) and additional *pop subgenre categories* (*soul*, *classical*, *synth-pop*, *r&b*). A third hierarchy level appears with *death metal* as a *subgenre category* of *metal* and *power metal* as a *subgenre category* of *heavy metal*. COMGET-G32 adds *house* as an *electronic subgenre category* and additional *rock* and *pop branches* (*blues*, *country*, *new wave*, *soft rock*, *dance-pop*, *reggae*). *hip hop subgenre categories* appear only in more fine-grained solutions (*pop rap* as 33rd genre, *trap* as 38th genre).

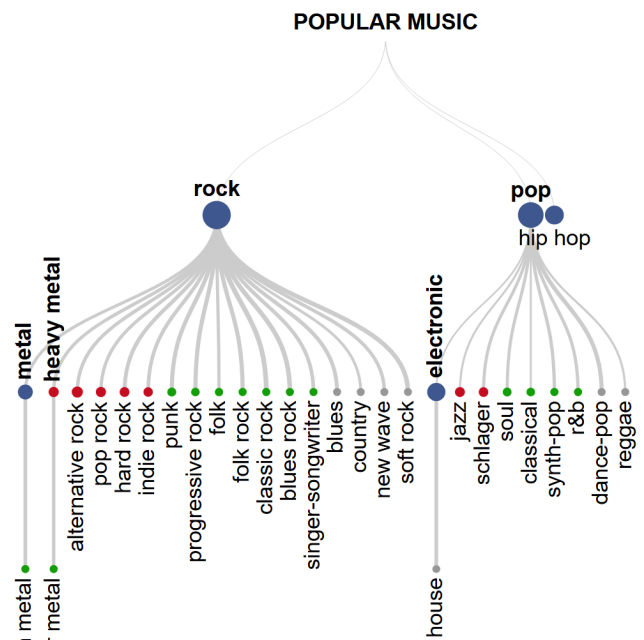


Figure 3: Folded COMGET-G hierarchies for POPTRAG corpus at four different granularities (5, 12, 25, 32 *genre categories*). Colors indicate additional categories introduced at each finer levels of granularity (blue = 5; red = 12; green = 25; grey = 32). Node size represents approximate track count in POPTRAG (sum of vote-weighted assignment probabilities).

4. RQ3: External Validity and ML-Classifier

4.1 Methods: Single Labels, Content Validity, Features, Workflow

To compare the COMGET-G genre assignments to other assignments frequently used in MIR and computational musicology and for training a classifier, each track in the corpus obtained a *single COMGET-G genre label* via majority voting: all valid original *MusicBrainz* tags assigned to a track were mapped to their *category assignment* in folded COMGET-G, then each track received the *category tag* with the highest summed votes. Ties were resolved by selecting

the *genre category* with lower total assignment probability in the full dataset, thereby favoring more detailed *genre category* assignments. We derived *single genre labels* for each track at four levels of granularity (COMGET-G5, COMGET-G12, COMGET-G25, COMGET-G32). For demonstrating the content validity of our solutions, we compared resulting *genre category assignments* with comparable granularity to genres labels assigned by the predicted *Rosamerica-Genres* from *AcousticBrainz* (Alonso-Jiménez et al., 2020), community assignments from the *Discogs* (2024) folksonomy, and genres retrieved from the API of the music streaming platform *Deezer* (nd). We analyzed similarity using contingency tables.

To determine if an automated classifier could predict the COMGET-G *genre categories* identified by FOLDCAT, we trained independent classifiers for each. To that end, we augmented POPTRAG with 192 descriptor variables covering *musical style*, *release context*, *lyrical content*, and *popularity* from music platforms *Spotify*, *MusicBrainz*, *AcousticBrainz*, *Genius*, and *Deezer* via ISRC matching or fuzzy string matching via either API-access or web scraping. *AcousticBrainz* provided community-calculated *audio features*; *MusicBrainz* supplied artist demographics (*persona*, *origin*, *birth year*, *gender*); *Spotify* and *Deezer* contributed *popularity* metrics and *Echo-Nest audio features*. From *Spotify's artist-level genre* tags we derived 69 *distribution-genre* flags (e.g., whether a track's artist is tagged as 'reggae' on *Spotify*) by applying the same co-occurrence algorithm used to build COMGET-G but operating on *Spotify* artist genres. *Spotify's distribution genre* categories can be a valuable signal of release milieu and test whether industry framing influences how listeners ultimately categorize music.

In our workflow, we first discarded observations with more than 40% missing values on any variable, retaining 155,093 tracks (92%). We then performed an 80/20 train–holdout split at the artist level to mitigate artist leakage effects (Flexer and Schnitzer, 2010). For training, we employed 5-fold cross-validation (also group-split by artist) for hyperparameter tuning. To stabilize categorical predictors, we drew a 20% subsample of the training set, created artist-grouped CV folds on that subsample, and relabeled any factor level that occurred fewer than 100 times in at least one fold as “other”. Predictors that collapsed to a single “other”-level as well as two-level predictors where the “other” level had fewer than 100 observations, were removed. This cleaning reduced the number of variables from 192 to 149. All remaining categorical predictors were dummy-coded, using the most common level as the reference.

Preprocessing was applied in sequence within each single fold: median imputation, adaptive class balancing (excluded for holdout set), and normalization. Adaptive balancing used a single tunable *target-ratio*

parameter governing rebalancing degree. The algorithm computed a reference count from outcome class frequencies (median for ≤ 10 classes; 60th percentile for 11–20 classes; 65th percentile for > 20 classes), then downsampled classes exceeding $\text{reference} \times \text{target ratio}$ using NearMiss (Zhang and Mani, 2003) and upsampled classes below $\text{reference} \div \text{target ratio}$ using Borderline SMOTE (Han et al., 2005) or regular SMOTE when sufficient danger observations were unavailable. This approach preserved more data than fixed-ratio sampling while remaining tunable: *target ratio* = 1 enforced perfect balance; higher values retained original imbalance. The reference-count heuristic scaled to problem complexity, preventing excessive downsampling at higher resolutions.

For hyperparameter search, we employed Bayesian optimization (Snoek et al., 2012) with a Latin hypercube initial grid (Huslage et al., 2011), early stopping if no improvement occurred after 10 iterations, and exploration of uncertain hyperparameter regions after 5 iterations without improvement. The best hyperparameters were selected using the one-standard-error rule (Hastie et al., 2009). We tuned hyperparameters across five folds using the macro-averaged F1 score (mean per-class F1, set to 0 for unpredicted classes) to treat all *genres* equally during evaluation.

Initially, we evaluated four different supervised ML algorithms based on a 20% subsample of the training data at low and medium granularity to identify the most suitable learner: elastic net (GLMNET; Tay et al., 2023), regularized discriminant analysis (RDA; Weihs et al., 2005), random forest (RF; Wright and Ziegler, 2017), and light gradient boosted machine (LightGBM; Shi et al., 2025). For each we tuned key hyperparameters: GLMNET — *penalty* and *mixture* in $[0, 1]$; RDA — *covariance fraction* and *identity fraction* in $[0, 1]$; RF — *feature fraction considered at each split* in $[1, \# \text{ of predictors}]$ and *leaf node size* in $[1, 100]$; number of trees was set to 500 for tuning and 1000 for final models; LightGBM — *boosting rounds* in $[50, 2000]$, *learning rate* in $[0.001, 0.1]$, *leaf node size* in $[1, 100]$, *tree depth* in $[3, 15]$, and *feature fraction* in $[1, \# \text{ of predictors}]$. For all learners, the target ratio used for adaptive balancing was tuned in $[1, \max(\text{observations per class})/\min(\text{observations per class})]$. For choosing the best learner, we employed a coarse hyperparameter search with a 10-level initial grid and a maximum of 25 iterations for Bayesian optimization.

After selecting the learner with best performance across folds, we performed a more extensive hyperparameter search with a 20-level initial grid and a maximum of 50 iterations with 5-fold crossvalidation on the full training set for all levels of granularity, trained the final classifiers on the full training set, and evaluated results on the holdout set. *Community certainty* served as a reference point for evaluating final classifier performance, since ML models cannot exceed the con-

sistency of the underlying human community assignments. For this comparison *certainty* was computed as the mean assignment probability across votes for each track assigned to the *single-label genre*. All random operations used the seed 42 for reproducibility.

4.2 Results: Comparison against external assignments

The *Rosamerica* genre set comprises eight categories, making it most comparable to COMGET-G5; *AcousticBrainz* lacked entries for 28% of POPTRAG tracks, introducing a modest popularity bias. The contingency table (Figure 4a) shows the strongest alignment between COMGET *metal* and *Rosamerica Rock* (83%). Other COMGET-G5 categories are distributed across multiple *Rosamerica* genres: COMGET *rock* mostly maps to *Rock* (45%), *Pop* (21%), and *Rhythm and Blues* (17%); *pop* maps primarily to *Pop* (35%) and *Rhythm and Blues* (29%); *electronic* maps to *Dance* (28%), *Rhythm and Blues* (21%), and *Pop* (19%), with a smaller share in *Hiphop* (14%); and *hip hop* maps to *Hiphop* (50%) and *Rhythm and Blues* (29%). Except for *metal*, all COMGET-G5 categories overlap with *Rhythm and Blues*, suggesting shared sound attributes in some tracks within these categories and a distinctive sound of *metal*. Given the broad scope of COMGET-G5 and the subgenres composing its categories, these mappings appear reasonable. The balance for *Rosamerica* assignments is marginally better than for COMGET-G5 assignments ($Gini_{COMGET-G5} = 0.488$, $Gini_{Rosamerica} = 0.417$).

Discogs' 15-genre taxonomy approximates COMGET-G12 in scale. For approximately 5% of tracks with COMGET-G assignments, we found no *Discogs* counterpart, introducing minimal popularity bias. The contingency table (Figure 4b) reveals strong alignment ($> 80\%$) for all COMGET-G12 categories with a single *Discogs* genre, except for *pop rock* (max. 77%), *shlager* (max. 76%), *jazz* (max. 71%), and *pop* (max. 30%). Even where *Discogs* includes a *Jazz* category, COMGET-G's *jazz* overlaps substantially with *Discogs'* *Electronic* (11%) and *Pop* (8%). Similarly, *pop rock* is distributed across *Discogs'* *Rock* (77%), *Pop* (12%), and *Electronic* (8%); *schlager* emerges in *Pop* (76%, similar label to its COMGET-G *supergenre*) with an overlap to *Rock* (16%). Five *Discogs* categories—*Blues*, *Brass & Military*, *Children's*, *Non-music*, and *Stage & Screen*—show minimal correspondence to any COMGET-G12 genre category, mostly consistent with our noise-reduction filtering and focus on popular music. COMGET-G's *pop* category is distributed widely across *Discogs* genres (*Electronic* 30%, *Pop* 20%, *Rock* 22%, *Funk / Soul* 10%), reflecting pop's eclectic character. Seven of twelve COMGET-G12 genre categories overlap most strongly with *Discogs'* *Rock*, suggesting *Discogs* lacks granular rock subgenre category labels needed to represent German popular music adequately. At this medium granularity, COMGET cate-

gory assignments show much a much more balanced distribution for POPTRAG than *Discogs* assignments ($Gini_{COMGET-G12} = 0.540$, $Gini_{Discogs} = 0.749$).

We found no *Deezer* assignment for 9% of POPTRAG tracks, while 26 *Deezer* genres appear in the corpus; the contingency table (appendix Figure B.1) shows $> 80\%$ alignment only for *blues rock* $\rightarrow$ *Rock* and *hip hop* $\rightarrow$ *Rap/Hip Hop*. Most COMGET categories distribute across two or three *Deezer* labels (10% threshold). In contrast, *electronic* maps to *Electro* (34%), *Pop* (24%), *Dance* (20%), and *Alternative* (12%). Also, *folk* maps to *Pop* (29%), *Alternative* (21%), *Rock* (13%) and *Folk* (12%). Notably, *pop* aligns strongly with *Deezer's* *Pop* category (68%). COMGET-G25 remains more balanced than *Deezer* ($Gini_{COMGET-G25} = 0.602$, $Gini_{Deezer} = 0.753$).

4.3 Results: Classification Performance for different degrees of granularity

Learner comparison (Figure 5) shows that all four learners perform within a similar range on the 20% subsample (mean $F1_{macro}$; CV = 0.717 – 0.758 for COMGET-G5 and $F1_{macro}$; CV = 0.468 – 0.503 for COMGET-G12). However, LightGBM distinctively emerged as the strongest performer at both granularities (COMGET-G5 and COMGET-G12) and was thus selected as the final learner for all subsequent analyses. While GLMNET ranked second best for COMGET-G5, both GLMNET and RDA performed worse for COMGET-G12 compared to tree-based methods, suggesting that models limited to primarily linear or low-order quadratic decision boundaries struggled to capture the increased complexity at finer outcome granularity.

The final GBM models on the full dataset (Table 2) achieved $F1_{macro} = 0.764$ (SE: 0.004) on CV folds and 0.743 on the hold-out set for COMGET-G5, indicating slight overfitting. Per-class F1 correlates strongly with *community assignment certainty* ($Cor_{Kendall} = 0.949$). Given mean *community certainty* of 90%, the classifier performs satisfactorily. Cohen's $\kappa = 0.674$ on the hold-out set ($\kappa_{CV} = 0.695 \pm 0.007$) is very close to established inter-rater agreement standards ($\kappa = 0.7$) in the social sciences (Hildebrandt et al., 2015, p. 327). The confusion matrix (Figure 6a) reveals the model struggles most with *electronic* (56% true positive-TP), often confusing it with *pop* (21%) and *rock* (19%). *Electronic* music's stylistic elements frequently appear across pop and rock subgenres, explaining the observed misclassifications. Similarly, *metal* (61% TP) misclassifications with *rock* (37%) reflect their close relationship.

COMGET-G12 performance degraded to $F1_{macro}$; CV = 0.519 ± 0.007 and $F1_{macro}$; hold-out = 0.544 ($\kappa_{CV} = 0.546 \pm 0.010$; $\kappa_{hold-out} = 0.551$), with lower *community certainty* (67%), indicating greater difficulty for both community and classifier. Per-class F1 and *certainty* remain highly correlated ($Cor_{Kendall} = 0.721$).

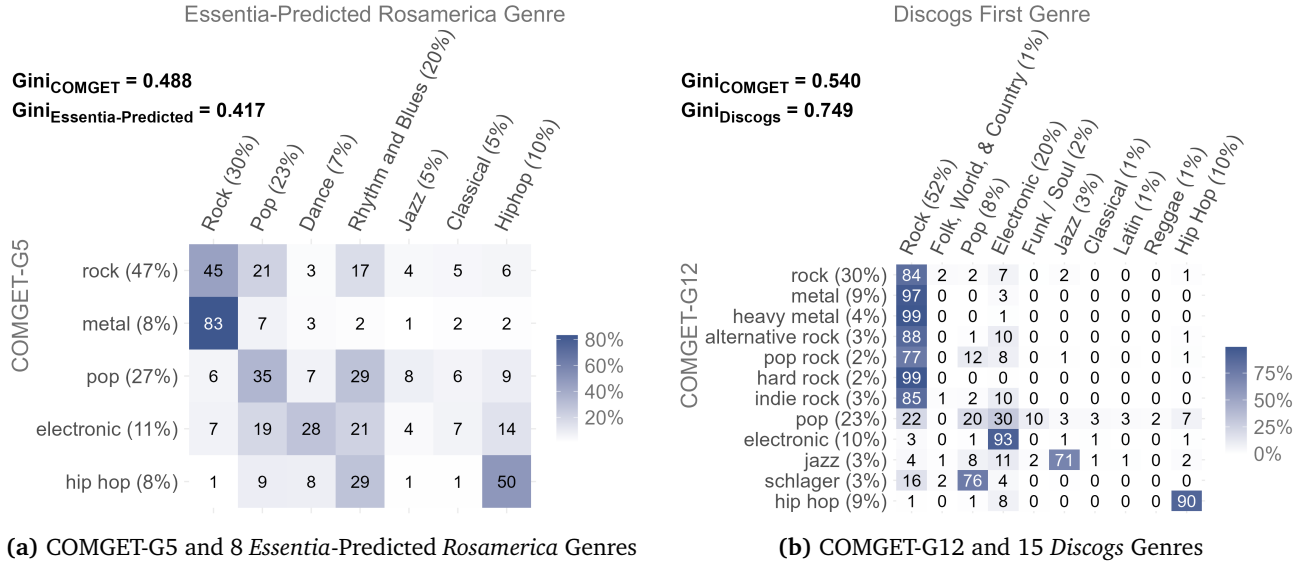


Figure 4: Contingency tables comparing COMGET + FOLDCAT assignments to external genre assignments commonly used in MIR and cultural research; values show $P(\text{External} | \text{COMGET})$. Discarded external genres for visualization with less than 2% relative frequency (*Rosamerica*: *Speech*, *Discogs*: *Blues*, *Brass & Military*, *Children's*, *Non-Music*, *Stage & Screen*).

# genres	boosting rounds [50, 2000]	tree depth [3, 15]	feature fraction [1, # of predictors]	leaf size [1, 100]	learning rate [0.001, 0.1]	target ratio [1, max/min]
5	93	12	68	46	0.072	4.815
12	1145	10	5	28	0.046	2.116
25	1512	9	85	91	0.005	2.641
32	176	9	55	15	0.0182	4.589

Table 2: Hyperparameter settings for final LightGBM models at different granularity levels.

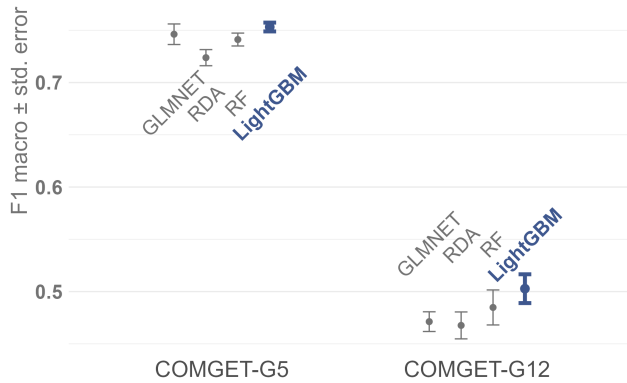


Figure 5: Performance comparison of different learners (GLMNET, RDA, RF, GBM) on 20% of the data set for low and medium genre granularity; plots show mean and standard error of $F1_{\text{macro}}$ across CV folds.

The confusion matrix (Figure 6b) shows primary confusion among *rock subgenres* (*alternative rock*, *hard rock*, *indie rock*, *pop rock*). *Heavy metal* (56% TP) is frequently confused with *metal* (34%) but less often vice versa (*metal* 71% TP, 13% confusion with *metal*), and sometimes with *hard rock* (10% of *hard rock* misclassified as *heavy metal*), despite being recognized as distinct from *rock* (3% confusion). *indie rock* (49% TP) and *jazz* (61% TP) are often misclassified as *electronic* (*indie rock* 13%, *jazz* 11%); *jazz* is also confused with *pop* (14%). *pop rock* performs worst ($F1 = 0.182$), matching its low *community certainty* (50%). It is primarily confused with high-certainty genres *rock* (22%) and *pop* (34%), suggesting *pop rock* shares ambiguous characteristics with both. Conversely, *schlager* and *hip hop* tracks are classified quite well ($F1 = 0.837$ and 0.858 , respectively) despite differing *community certainties* (67% and 100%), indicating these *genres* possess very distinct characteristics in the feature space.

COMGET-G25 showed further loss in classification performance ($F1_{\text{macro}; \text{CV}} = 0.354 \pm 0.006$; $F1_{\text{macro}; \text{hold-out}} = 0.348$; $\kappa_{\text{CV}} = 0.436 \pm 0.009$; $\kappa_{\text{hold-out}} = 0.430$) alongside decreased *community certainty* (56%). The correlation between *certainty* and $F1$ weakened ($\text{Cor}_{\text{Kendall}} = 0.589$), yet remained high, still suggesting other factors influence performance at higher granularity. The confusion matrix (see appendix figure B.2a) again reveals persistent *rock subgenre* confusion (*alternative rock*, *blues rock*, *classic rock*, *folk rock*, *pop rock*) and *metal*-related confusion (*death metal*, *folk rock*, *hard rock*, *heavy metal*, *power metal*). A secondary confusion center emerges around *pop*, which frequently misclassifies with *alternative rock*, *classical*, *electronic*, *folk rock*, *pop rock*, *r&b*, *singer-songwriter*, *soul*, and *synth-pop*. The model performs well ($F1 > 0.6$) on *classical*, *electronic*, *hip hop*, *metal*—genres with distinct feature signatures and moderate to high com-

munity certainty (ranging 67%-100%). Poor performers ($F1 < 0.2$; *alternative rock*, *blues rock*, *classic rock*, *folk*, *folk rock*, *pop rock*, *power metal*, certainty ranging 33%-50%) represent fuzzy genres difficult to classify for both community and model.

Consistent with decreased community certainty (54%), the model for COMGET-G32 showed a lower classification performance ($F1_{\text{macro}; \text{CV}} = 0.344 \pm 0.003$; $F1_{\text{macro}; \text{hold-out}} = 0.332$; $\kappa_{\text{CV}} = 0.392 \pm 0.005$; $\kappa_{\text{hold-out}} = 0.339$, see appendix figure B.2b). Also, the correlation between *certainty* and $F1$ declined further ($\text{Cor}_{\text{Kendall}} = 0.560$), indicating *community uncertainty*. However, correlation is still high. Given similarity to COMGET-G25, please refer to the appendix for a detailed confusion matrix.

Figure 7 shows the top 20 most important features for automated classification of COMGET-G5 and COMGET-G12 with the final LightGBM learner (for COMGET-G25 and COMGET-G32 see appendix figure B.2). While we find similar features in the top 20 across all three levels of granularity (e.g., *acousticness* (Echo-Nest), *artist distributed as “metal”*, *age of artist*), the importance of single features differs. For COMGET-G5, features that describe general style characteristics (e.g., *acousticness*, *speechiness*, *explicitness of lyrics*) and artist persona (e.g., *single artist vs. band*, *age of artist*) are most important. For COMGET-G12, *popularity* features and the *release year* gain importance. For COMGET-G25 and COMGET-G32 the most important features are mostly artist-related, indicating that fine-grained genre distinctions rely strongly on release-milieu signals rather than audio properties.

5. Discussion

With the present article, we introduce a novel and methodologically transparent procedure for constructing well-balanced hierarchical genre taxonomies and according genre category assignments from listener-community discourse contained in folksonomies on open music databases. Specifically, we make three key contributions: (1) an algorithm to construct a balanced hierarchical genre tree (COMGET) from folksonomy genre tag co-occurrences, (2) a folding algorithm (FOLDCAT) that generates optimally balanced hierarchical genre categories and final track-assignments at user-specified granularities from such trees; finally, we (3) demonstrate that resulting community-based genre assignments are a) comparable to existing folksonomy genre taxonomies (i.e. *Discogs*) and b) can be learned well with supervised machine learning by drawing on openly available music descriptors. Hence, our approach can be used as a transparent, reproducible, and empirically grounded alternative to commercial or expert-curated genre taxonomies in future applied or basic research. We demonstrated our approach with a large German popular music corpus (POPTRAG), drawing on *MusicBrainz* folksonomy

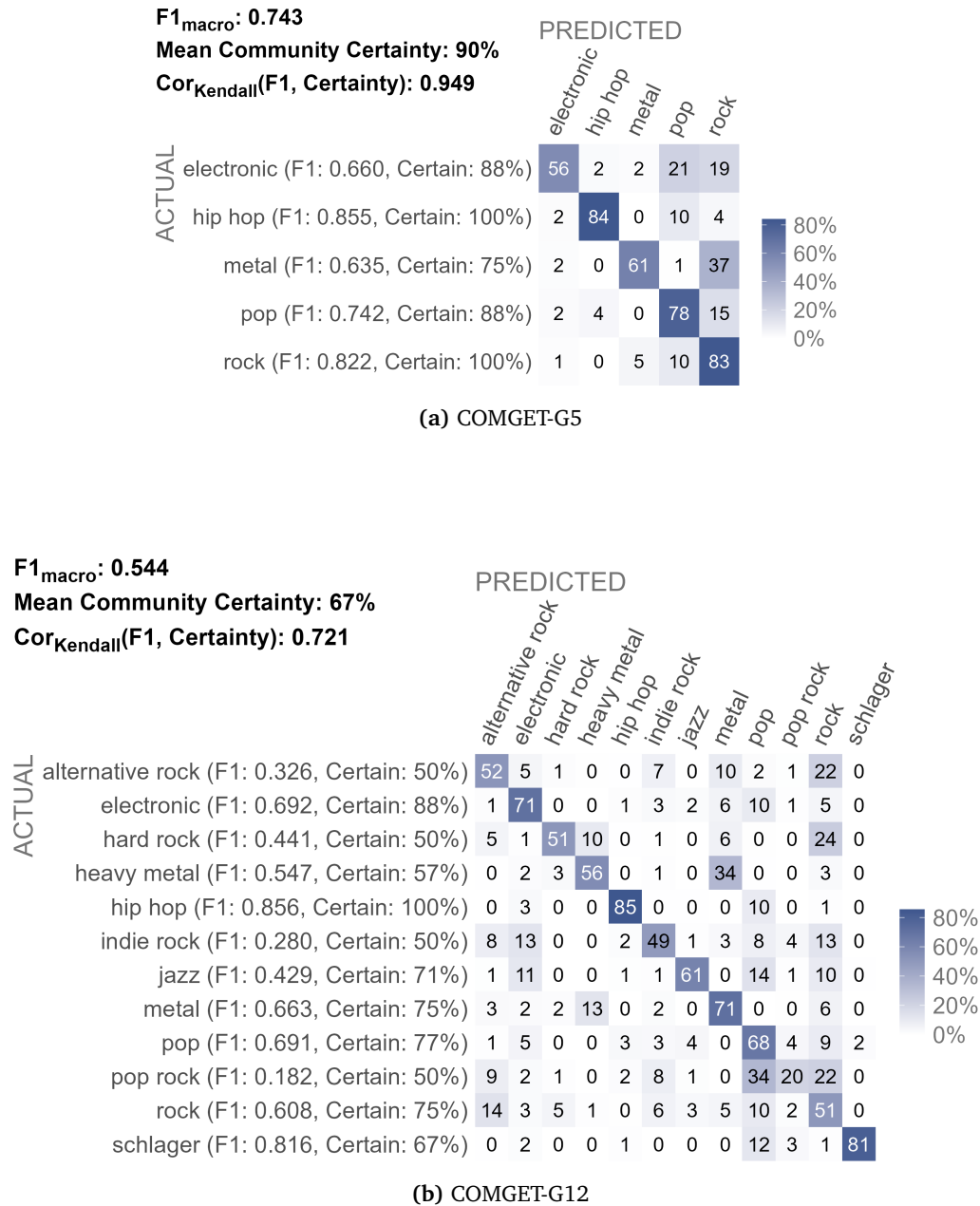


Figure 6: Confusion matrices for the final GBM models on the hold-out set. Values show $P(\text{Predicted}|\text{Actual})$; *certainty* indicates the mean assignment probability for each *genre category* across community votes.

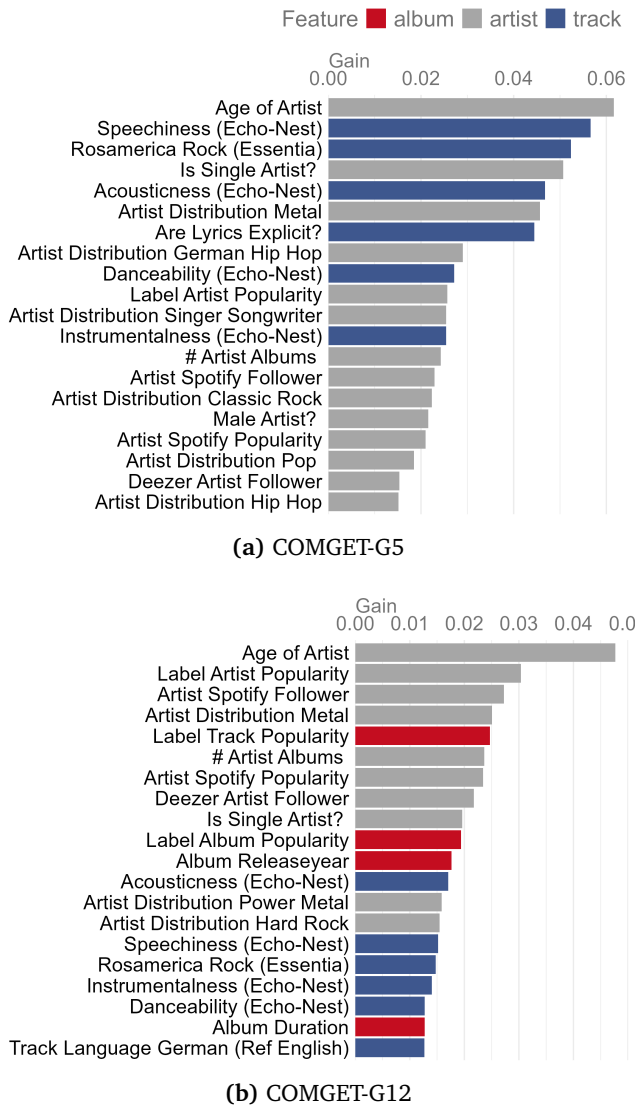


Figure 7: Top 20 most important features for automated classification at low and medium granularity. Values show relative feature importance from the final LightGBM models.

votes. Nevertheless, the method can in principle be adopted to any other music corpus and other genre tagging systems that provide co-occurrences of tags (e.g. *Spotify*, *YouTube*, *SoundCloud*).

5.1 Benefits for MIR Research and Applications

A core methodological challenge in MIR stems from genre recognition benchmarks that lack explicit theoretical grounding (Green et al., 2024). Our approach operationalizes genre-as-discourse (Foucault, 2013; Neale, 1980), treating classification as an empirical question rather than an ontological given. This connects computational methods to popular music studies and enables MIR to engage with humanities scholarship on cultural categorization (DiMaggio, 1987).

Furthermore, our single-folksonomy approach ensures epistemic consistency. Mixed-source methods conflate different communities' genre concepts across platforms with varying demographics, tag voting systems, tag pools, and discursive contexts, introducing uncontrolled variance (Schreiber, 2015; Bogdanov et al., 2019). Single-source taxonomies eliminate cross-platform variance, ensuring all genre assignments reflect a single community's discourse rather than mixing incompatible categorization systems. Combined with no arbitrary thresholds throughout the pipeline and open-source implementation, this ensures reproducibility and portability to any multi-tag corpus, as demonstrated with *Spotify* distribution genres. Researchers can replicate the method on their target population's folksonomy, generating taxonomies specific to their chosen listener community.

FOLDCAT addresses diverse practical needs while enabling cross-study comparison. The folding algorithm generates optimally balanced genre categories at multiple granularities using a single statistical criterion (Gini coefficient) across hierarchy levels. In addition, the underlying genre category assignments stay probabilistic. Hence, the framework quantifies achievable performance ceilings through community certainty, since models cannot exceed human consistency (Sturm and Flexer, 2023). Strong correlations between certainty and classifier performance validate this approach, grounding MIR in listener perception limits. Classifier performance also demonstrates general learnability of the identified genre categories, while systematic confusion patterns reveal fuzzy boundaries (*rock* subgenres), stylistic permeability (*electronic* across *pop/jazz/indie*), and hierarchical nuance (*heavy metal* versus *metal* asymmetry). These patterns inform which granularities suit different research questions and reaffirm that genre labels remain learnable from the dimensions suggested by the anonymized authors.

5.2 Benefits for Computational Musicology and Cultural Analysis

COMGET and FOLDCAT consequently implement power-critical concepts from popular music studies and cultural analysis (du Gay et al., 2013) into computational methods for the application on large popular music corpora. Our approach employs democratic aggregation throughout: community votes determine co-occurrences, strongest links define hierarchies, majority voting assigns single labels. Resulting community genre taxonomies, genre categories, and genre category assignments reflect collective hierarchical listener discourse rather than industry or expert impositions, democratizing genre knowledge production for computational musicology. This approach can improve the validity of music-preference studies by aligning measurements with how listeners actually categorize music.

Crucially, COMGET taxonomies capture how listeners organize popular music genres in contemporary reception rather than musicological historical genealogies (Pachet and Cazaly, 2000). The POPTRAG demonstration reveals unambiguous genealogical links (*rock* → *alternative rock*) alongside counterintuitive ones that invite cultural interpretation rather than genealogical correction. For instance, *classical* and *jazz* emerge as *pop* subgenres. Two factors explain this placement: POPTRAG mostly contains accessible and crossover works—popular opera arias, smooth jazz compilations, chart-oriented classical crossover artists. This corpus profile combines with listeners' tendency to apply broad labels when tagging popular music (Bogdanov et al., 2019). Similarly, both *heavy metal* and *metal* appear as *rock* subgenres, reflecting shared historical roots in 1970s psychedelic and hard rock while acknowledging *metal's* later distinctiveness (Gracyk, 2016).

Our POPTRAG analysis illustrates the method's capacity to reveal genre discourse structures computationally. *Rock*, *pop*, and *hip hop* emerged as three primary clusters for German popular music genres in Germany, partially aligning with Silver et al. (2016) who identified three main international clusters analyzing MySpace artist networks in 2007. However, structures differ: *rock* forms a distinct cluster in POPTRAG versus Silver et al.'s scattered "Niche" genres; *hip hop* shows minimal differentiation (consistent with Silver et al.); *pop* exhibits stylistic diversity versus Silver et al.'s "Rock 'n Roll" category. Direct comparison remains limited—Silver analyzed artists on a social networking platform, we analyze tracks in a national listening corpus across different decades—yet the method enables such comparative analyses across corpora, revealing how genre discourse varies by platform, national context, and historical period (Pichl et al., 2017; Nie, 2022).

Transparent automated classifier analysis reveals

how listener communities perceive genres: artist-level features dominate predictions across all granularities in our German popular music analysis, with distribution genre flags (*Spotify* artist tags) and artist demographics outweighing audio features—a finding consistent with genre theory emphasizing release milieu (Neale, 1980). Stylistic features play a secondary role: audio features distinguish broad categories, but bag-of-words and lexical lyric features prove weak, pointing toward LLM-based embeddings as a promising direction (Prevedello et al., 2024). Popularity gains importance at finer resolutions as a genre signal itself (Holt, 2007). Genre-specific confusion patterns further illuminate boundary strength: e.g., *rock* subgenre overlap (*alternative*, *pop rock*, *indie*) reveals fuzzy boundaries; *electronic* permeability across *pop*, *jazz*, and *indie* reflects stylistic fluidity; and *schlager-hip hop* separation demonstrates distinct release milieus and stylistic features.

Beyond cultural research, this method offers new possibilities for stakeholders across the music industry: radio programmers and playlist editors to gain insight into how audiences organize music; A&R teams to identify gaps between industry positioning and listener reception; marketing departments to recognize which subcultural boundaries audiences perceive versus industry-imposed categories; music journalists to ground critical language in empirical listener consensus; and platform designers to align recommendation logic with listener mental models.

5.3 Limitations and Future Directions

Several constraints limit interpretation and generalization of our exemplary results. MusicBrainz launched in 2000 and does not provide individual vote timestamps, so COMGET-G aggregates roughly 25 years of listener votes spanning a period when genre boundaries and perception likely shifted. Platform user demographics and expertise remain opaque (approximately 1.2 million anonymous editor accounts; MusicBrainz, nd), preventing assessment of demographic or expertise bias. For the analysis we excluded 36% of tracks with insufficient votes, thereby introducing slight popularity bias—acceptable for our popular music focus but possibly problematic for niche genres. Methodologically, collinearity among predictors complicates feature importance interpretation. Single-label reduction loses multi-tag nuance, though this simplification proves necessary for many applied research tasks and aligns with our democratic approach.

Future work should pursue three directions. First, convene popular music experts to review the taxonomy, label exemplar tracks for genre categories at different levels of granularity, and identify misassignments to strengthen COMGET-G's validity. Second, replicate the pipeline on independent folksonomies and regional corpora of tracks (e.g., US and UK charts)

to reveal how national listening cultures reshape genre structure. Third, enhance classifier prediction through advanced lyric features or learners.

Competing Interests

The authors have no competing interests to declare.

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